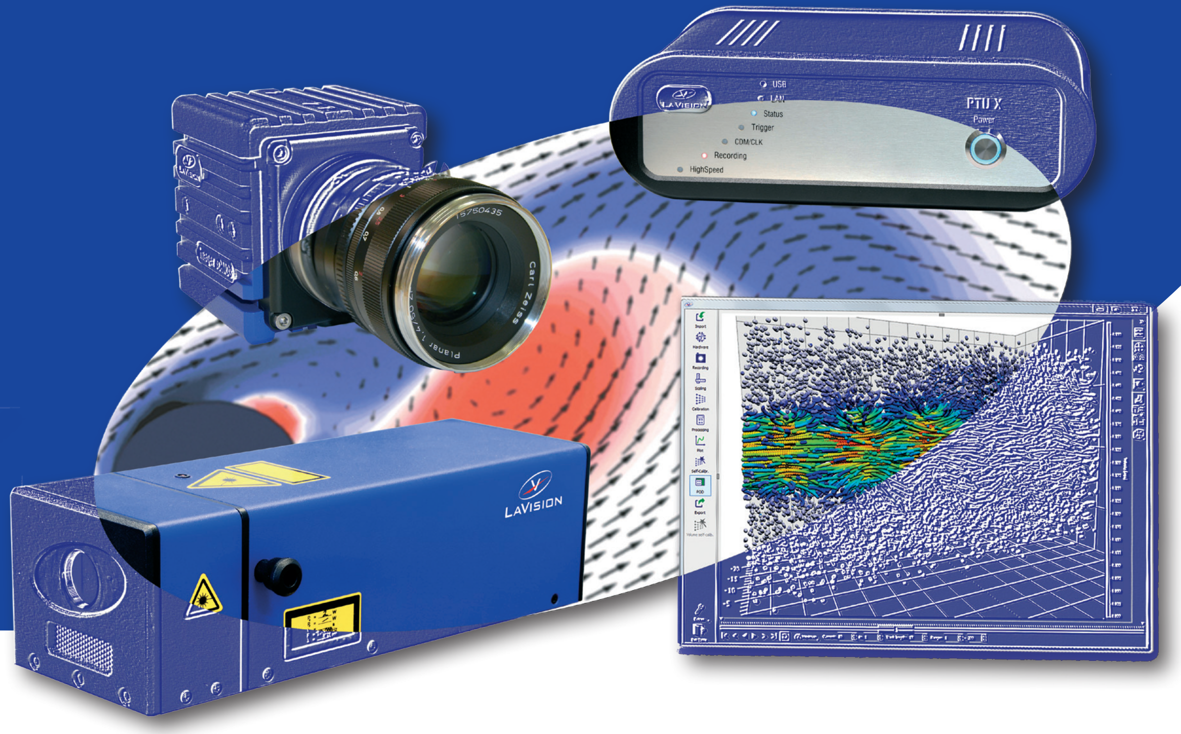

Product Manual

Sheet Optics (divergent)

Item Number(s): 1108405, 1108516, 1108406

LAVISION
FOCUS ON IMAGING

Product Manual for **DaVis 10**

LaVision GmbH, Anna-Vandenhoeck-Ring 19, D-37081 Göttingen
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Contents

1 Safety Precautions

Before working with your **LaVision** system, we recommend to read the following safety precautions. Observing these instructions helps to avoid danger, to reduce repair costs and downtimes, and to increase the reliability and life of your **LaVision** system.

1.1 Laser Safety

If a laser[1] is integrated in your system, it is important that every person working with it has fully read and understood these safety precautions **and** the laser manual of the specific laser/LED.

Lasers included in **LaVision** systems may belong to Class 4 laser devices, which are capable of emitting levels of both visible and invisible radiation that can cause damage to the eyes and skin. It is absolutely necessary that protective eyewear with a sufficiently high optical density be worn at any time when operating the laser. The goggles must protect against all wavelengths that can be emitted, including harmonics. See your laser´s manual for further details.

Class 4 laser beams are by definition a safety and fire hazard. The use of controls, adjustments or performance of procedures other than those specified in the **LaVision** manual and the laser manual may result in hazardous radiation exposure.

AVOID EYE AND SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION. FOLLOW THE INSTRUCTIONS YOU CAN FIND IN THE CORRESPONDING LASER MANUAL FOR PROPER INSTALLATION AND SAFE OPERATION. USE PROTECTIVE EYEWEAR ALL THE TIME WHEN OPERATING THE LASER.

[1] In the following, 'laser' means any kind of laser, in particular Nd:YAG and dye lasers as well as Optical Parametric Oscillators at any wavelength and output energy. Also for high-power LEDs precautions should be taken.

Important instructions for safe laser handling:

- Before operating the laser, contact your laser safety officer.
- Read and understand the instruction manual of the particular type of laser. Take special care with respect to laser emission, high voltage and hazardous gases if in use.
- Declare a controlled access area for laser operation. Limit access to trained people. Never operate the laser in a room where laser light can escape through windows or doors. If possible, cover beam paths to avoid obstacles getting into the beam.
- Provide adequate and proper laser safety goggles to **all** persons present who may be exposed to laser light. The selection of the goggles depends on the energy and the wavelength of the laser beam as well as on the operation conditions. Check the laser´s manual for a detailed description.
- While working with lasers do not wear reflective jewelry like watches and rings, as these might cause accidental hazardous reflections.
- Avoid looking at the output beam, even diffuse reflections can be dangerous.
- Operate the laser at the lowest beam intensity possible.
- Avoid blocking the output beam or reflections with any part of the body. Use beam dumps to avoid reflections from the target.
- Wear clothes and gloves which cover arms and hands to avoid skin damage when handling in the optical path. Especially UV radiation can cause skin cancer.

1.2 Seizures Warning

WARNING: HEALTH HAZARD! STROBE LIGHTING CAN TRIGGER SEIZURES! Some people (about 1 in 4000) may have seizures or blackouts triggered by flashing lights or patterns. This may occur when viewing stroboscopic lights or objects illuminated by such devices, even if a seizure has never been previously experienced. Anyone who has had a seizure, loss of awareness, or other symptoms linked to an epileptic condition should consult a doctor before operating systems which include flashing lights, strobe lights, or a pulsed or modulated laser.

Stop operating the system immediately and consult a doctor if you have one of the following symptoms:

- convulsions, eye or muscle twitching, loss of awareness, altered vision, involuntary movements, disorientation.

To reduce the likelihood of a seizure when operating a system:

- Do not look directly at flashing light sources or on illuminated objects, e.g. into a strobe light or a flashing LED panel.
- Operate the system in a well-lit room.
- Take frequent breaks in normally illuminated areas.

1.3 Camera / Image Intensifier Safety

The camera integrated in your system is based on a CCD (Charge Coupled Device) or CMOS (Complementary Metal-Oxide Semiconductor) sensor with high resolution and high sensitivity. Optionally your system is equipped with a built-in or external image intensifier.

A LASER BEAM FOCUSED ON THE CHIP OR INTENSIFIER, EITHER DIRECTLY OR BY REFLECTION, CAN CAUSE PERMANENT DAMAGE TO THE CHIP OR INTENSIFIER. ANY LASER POWERFUL ENOUGH TO PRODUCE LOCALIZED HEATING AT THE SURFACE OF THE CHIP OR INTENSIFIER WILL CAUSE DAMAGE EVEN WHEN THE CAMERA OR INTENSIFIER POWER IS OFF. A CHIP OR INTENSIFIER DAMAGED BY LASER LIGHT IS NOT COVERED BY ITS WARRANTY.

Important instructions for safe camera handling:

- Fully read and understand the instruction manual of the specific type of camera.
- Put the protection cap on the camera lens whenever you do not take images, especially when the laser beam is adjusted. Switching off the camera / image intensifier does not protect the chip from damage by laser light.
- Use full resolution of the sensor and always read out the complete chip to have control of the intensity on all areas of the sensor.

- Make sure that no parts of the image are saturated, i.e. the intensity is below maximum gray level ($<$ 4095 counts for a 12-bit camera, $<$ 65535 counts for a 16 bit camera, ...).
- Start measurements with the lowest laser power and a small aperture of the camera lens.
- Increase laser power step by step and check the intensity on the corresponding image. Make sure that the sensor does not run into saturation.
- Bright parts in the experiment, like reflections on walls or big particles, will limit the maximum laser power. Modify the optical arrangement of your setup in order to remove bright reflections from the camera image.

2 Sheet optics

The **LaVision** light sheet optics can be used for PIV, planar laser-induced fluorescence and flow visualization studies. The dimensions of the light sheet can be adjusted by refocusing the light-sheet thickness as well as by interchanging different divergent lenses. The light-sheet optics can be used together with high-power Nd:YAG lasers and handles beam diameters of up to 12 mm. The light sheet optics is compatible with **LaVision**'s light guide arm. The optics are integrated in protective housing and the modular design allows high flexibility.

Three different versions are available:

- 1108405 made with BK7 lenses with a high power anti-reflection coating for 532nm. See figure 2.15 for a detailed performance of this coating.
- 1108516 made with fused silica lenses with a triple center anti-reflection YAG-coating suitable for 266, 355 and 532nm. This coating is also suitable for all wavelengths between 266 and 532nm. See figure 2.16 for a detailed performance of this coating.
- 1108406 made with uncoated fused silica lenses for the whole UV to VIS (230nm-800nm) spectrum

See chapter2.10 for more specifications.

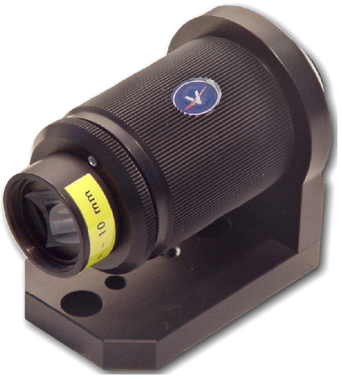

Figure 2.1: Sheet optics on base plate.

2.1 Safety Precautions

Serious eye and skin damage possible. Scattered or reflected light from the light sheet optics and the focused light from the exit can seriously damage your eyes and skin. Avoid direct exposure, even to reflections. Wear suitable protective goggles and gloves.

2.2 Accessories

Figure 2.2: Sheet optics accessories.

- **Sheet optics:** The sheet optics housing including two telescope lenses.
- **Two Caps:** For the sheet optics housing in order to avoid contamination of the lenses inside when the sheet optics is not used.
- **Two divergent lenses:** Two cylindrical lenses that form the laser beam to a light sheet. You may select between two different divergence lenses. The focus length will determine the aperture of the light sheet:
  $f = -10\,\text{mm}$ (large aperture) or $f = -20\,\text{mm}$ (small aperture). Article 1108406 and 1108516 contain a third divergent lens: $f = -50\,\text{mm}$. This lens will create an even smaller sheet height.

- **One laser head adapter:** May be used to fix the sheet optics body directly to a Quantel EverGreen laser head. See chapter 2.10 for a detailed sketch.
- **Base plate:** In case there is no possibility to fix the sheet optics directly at the laser head you can use the base plate to arrange the sheet optics independently. See chapter 2.10 for a detailed sketch.
- **Thorlabs–C-Mount Adapter:** In order to change from 1.035”-40 UNS Thorlabs thread to standard C-Mount.

2.3 Mounting

For small laser heads it is most convenient to fix the sheet optics housing at the laser head directly. The sheet optics housing can be fixed via 1.035”-40 UNS Thorlabs thread or with the adapter to a c-mount thread to laser head. If you have a Quantel EverGreen Laser you can use the laser head adapter that needs to be fixed at the laser head with 4 screws.

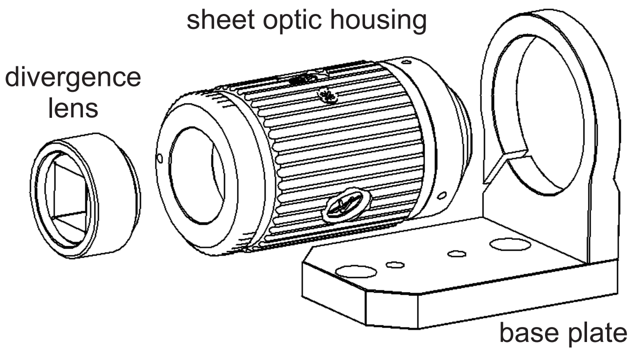

Figure 2.3: Sheet optics on base plate.

In case you need to fix the sheet optics independently you can use the base plate. Fix the sheet optics housing in the mounting bracket and use corresponding UNC- or M6 threads in the base plate to fix it to suitable supports.

2.4 Changing the Sheet Height

The height of the light sheet is mainly determined by the focal length of the cylindrical lens. In order to change the sheet height you need to change the cylindrical lens. **LaVision** offers cylindrical lenses with different focal length for different apertures (Article 1108405: $f = -20$ mm, $f = -10$ mm, article 1108516 and 1108406: $f = -20$ mm, $f = -10$ mm, $f = -50$ mm, on request: $f = -6$ mm, $f = -3$ mm). By the modular design the tube with the cylindrical lens can easily be removed and replaced by another one. To adjust the orientation of the light sheet, loosen the clamping screw on the front side of the sheet optics housing and turn the ring to the desired position.

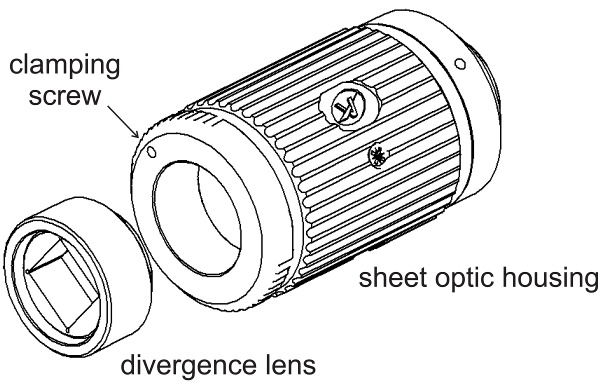

Figure 2.4: Changing the divergence angle.

2.5 Aperture Angle

Since the beam divergence of the laser beam itself is neglected, the aperture angle α of the light sheet is basically determined by the focal length f of the cylindrical lens and the beam diameter d at the lens position. According to the geometry in the figure below you find the relation $\tan(\alpha/2) = d/(2f)$.

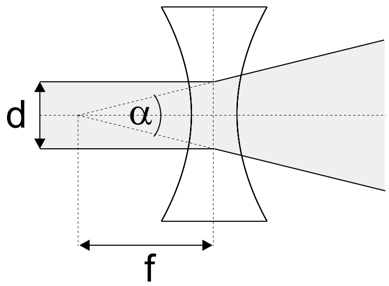

Figure 2.5: Relation between aperture angle α, focal length f and beam diameter d.

Example: The cylindrical lenses of article 1108405 have a focal length of $f = -10$ mm and $f = -20$ mm. For a beam diameter of $d = 4.5$ mm (Solo120) this will give an aperture angle of about $\alpha = 24°$ and $\alpha = 12°$.

2.6 Lens Configuration

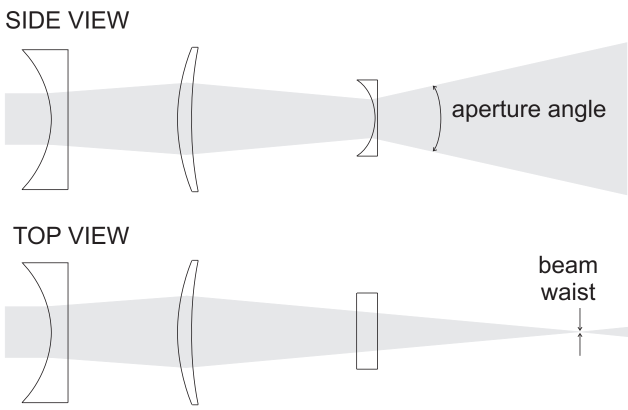

Figure 2.6: Lens configuration.

Basically, the light sheet is generated by one cylindrical diverging lens. Using a laser with a sufficiently small beam diameter and divergence, this cylindrical lens can generate an appropriate light sheet. For a laser with larger beam diameter and divergence, the divergence lens is combined with two telescope lenses to generate a thin light sheet. The telescope is used to focus the light to an appropriate thickness. The aperture and height of the light sheet is mainly given by the focal length of the cylindrical lens. The adaptation of the light sheet height has to be done by changing the cylindrical lens. The adjustment of the thickness of the light sheet can be controlled by the separation of the spherical lenses.

2.7 Changing the Focus Length

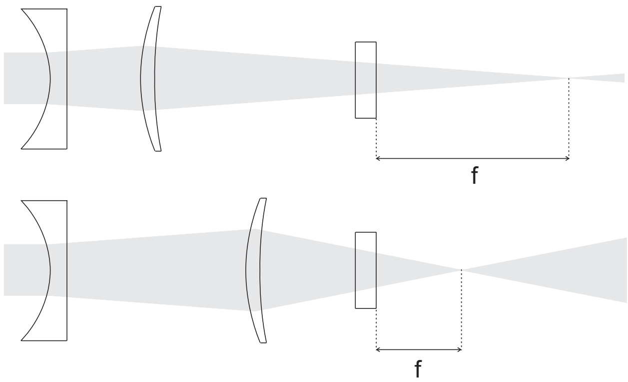

Figure 2.7: Sheet optics focusing.

The telescope lenses are arranged inside the sheet optics housing. By turning the focus ring the separation of the two spherical lenses can be changed in order to adjust the thickness of the light sheet. In case of a short separation of the lenses the resulting focal length of the sheet optics is approx. 2000 mm, with an enlarged separation between the spherical lenses the focal length of the sheet optics will be reduced to approx. 300 mm.

2.8 Adapter with 90° viewing direction

An adapter with 90° viewing direction (art. no. #1108407) is available for the standard sheet optics. It's only useable with laserlight in a wavelength of 532 nm.

The sheet optics has to be mounted at the outlet of the laserhead, followed by the adapter and the divergent lens. The tube at the adapter outlet has a long thread to adjust the sheet direction. Once the direction has been found the tube can be locked with a lock nut, figure 2.8.

Figure 2.8: Sheet optic with adapter and divergent lens.

The mirror inside the adapter is pre-adjusted. By loosening screws outside the cage the mirror including it's holder can be rotated - see fig. 2.9.

Figure 2.9: Rotate mirror.

By adjusting the screws at the front surface the mirror can be tilted. For tilting the mirror e.g. clockwise the right screw has to be screwed out and the left screw in, figure 2.10.

Figure 2.10: Tilt mirror.

2.9 Adapter for 90° viewing direction, article #1109201/ #1109727

Two adapters for 90° viewing direction (art. no. #1109201 and art. no. #1109727) are available for the **LaVision** sheet optics.

- Adapter #1109201 can be used with laser light in the wavelengths of 266 nm, 527 nm and 532 nm.
- Adapter #1109727 can be used with laser light in the wavelengths of 355 nm and 532 nm.

The sheet optics attaches to the outlet of the laser head. followed by the adapter for 90° viewing direction and the divergent lens. The tube at the adapter outlet has a long thread to adjust the orientation of the light sheet. Once the laser light sheet is in the desired orientation, the position can be locked by tightening a lock nut, figure 2.11.

Figure 2.11: #1109201/ #1109727 with divergence lens mounted at sheet optics

The position of the light sheet can be adjusted by the adjustment screws, figure 2.12. The adjustment screw closer to the sheet optics body is moving the light sheet in the direction lateral to the incoming laser beam. The adjustment screw closer to the divergence lens is moving the light sheet in the direction of the incoming laser beam.

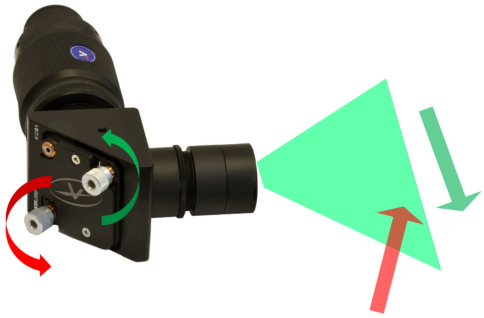

Figure 2.12: Adjusting sheet position

2.10 Specifications

- Sheet thickness: 0.5 mm - 2.5 mm (adjustable, but dependant on M2 and size of Laserbeam).
- Divergence lenses: $f = -20$ mm and -10 mm. 1108406 and 1108516 additionally with $f = -50$ mm.
- Divergence angle $\alpha = 10°$, $20°$, $3°$ @ $d = 3$ mm beam diameter.
- Working distance: 300 mm - 2000 mm.
- Coatings:
  - **1108405:** AR for 527/532 nm optimized for 532 nm with overall transmission $> 95\%$,
  - **1108516:** AR specifically made for 266, 355 **and** 532 nm. Also suitable for wavelengths inbetween these three.
  - **1108406:** without coating
- Dimensions: approx. Ø43 mm $\times$ 90 mm.
- Mounting:
  - directly to the laser head adapter via 1.035”-40 UNS Thorlabs thread
  - directly to the laser head via c-mountr adapter
  - directly to the laser head via Quantel EverGreen adapter
  - via baseplate and UNC- / M6 screw on e.g. posts and columns

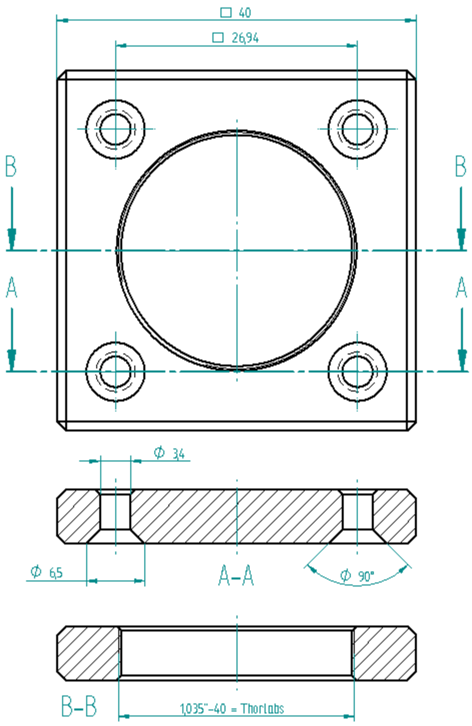

Figure 2.13: Quantel EverGreen laser head adapter with Thorlabs thread for SheetOptics.

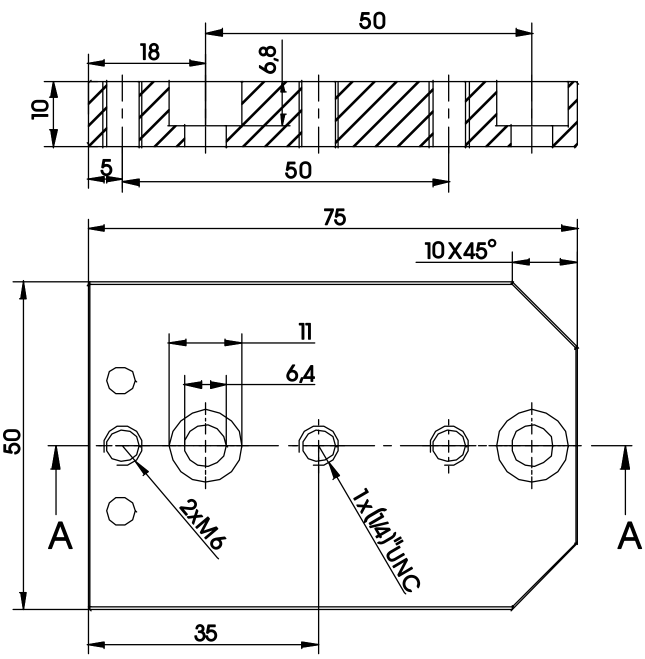

Figure 2.14: SheetOptics baseplate with 1/4 UNC / M6 threads for mounting.

2.10.1 Coatings and Damage thresholds

The damage threshold depends on your setup. Following values in table 2.1 are given in energy density (J/cm^2) for a gaussian shaped hotspot free laser beam. To compare with your laser energies scale your specific energy density to 1 cm^2. Keep in mind that forming (focussing, diverging) the laser beam changes the energy density. Consider puls lengths and avoid hotspots in your laser beam profile. Keep lenses absolute clean while in use. See Chapter 3 for cleaning and maintenance instructions.

Sheet Optics	**Lens type**	**damage threshold**
1108405	BK7 532nm AR-coated. See figure 2.15 for a detailed performance of this coating and 2.17 for a detailed transmission curve of BK7 substrate	2,0 J/cm^2 @ 5 ns
1108516	fused silica lenses with a YAG-coating suitable for 266-532 nm. See figure 2.16 for a detailed performance of this coating and 2.18 for a detailed transmission curve of fused silica substrate.	2,0 J/cm^2 @ 532 nm @ 10 ns 1,0 J/cm^2 @ 355 nm @ 10 ns 0,5 J/cm^2 @ 266 nm @ 10 ns
1108406	fused silica uncoated for the whole UV to VIS (230 nm - 800 nm) spectrum. See figure 2.18 for a detailed Transmission curve of fused silica substrate	fused silica like

Table 2.1: Overview of SheetOptics types and corresponding damage thresholds

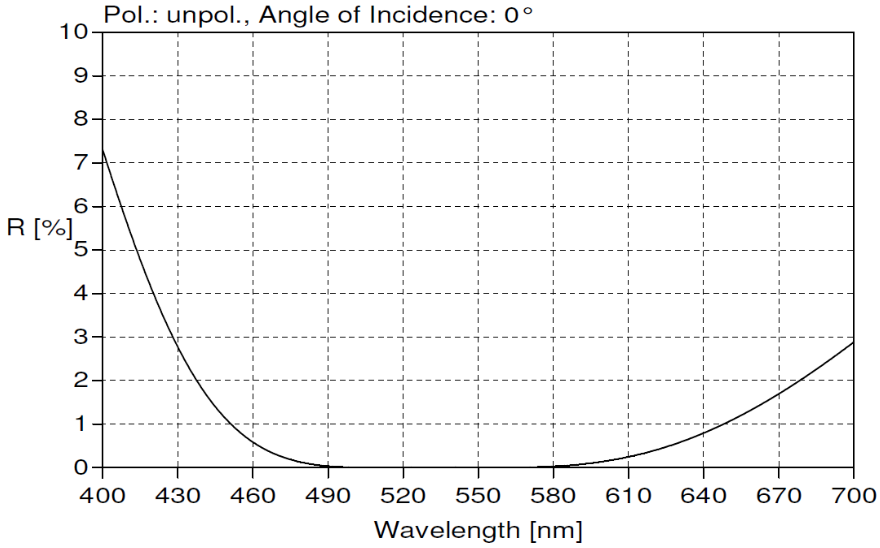

Figure 2.15: AR-coating of Sheetoptics 1108405 for 532/527 nm (VIS) wavelengths

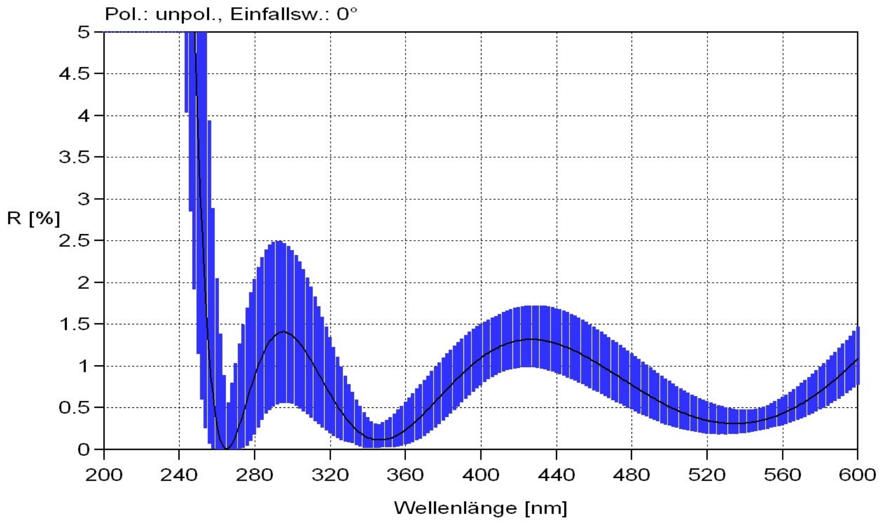

Figure 2.16: AR-coating of Sheetoptics 1108516 for YAG wavelengths

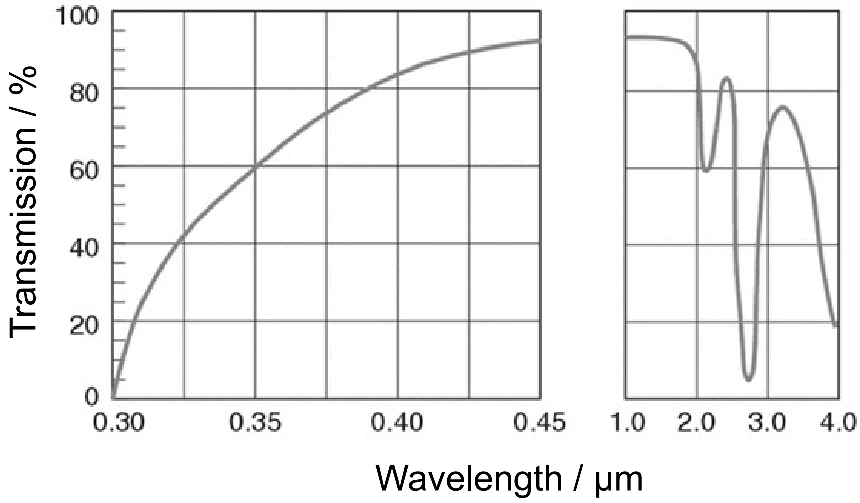

Figure 2.17: Typical transmission of BK7 substrate

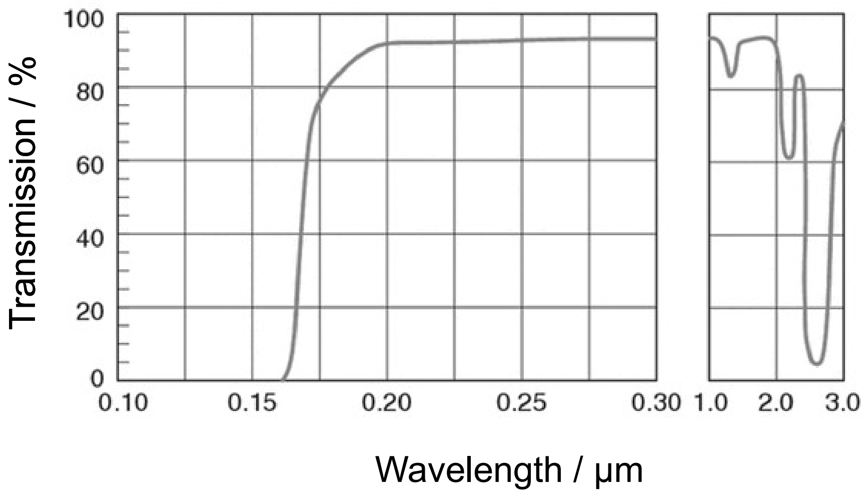

Figure 2.18: Typical transmission of fused silica substrate

3 Cleaning Optics

LaVision systems include devices which are equipped with delicate lenses and CCDs. To reduce repair costs and downtimes and to increase the reliability and lifetime of your system, it is extremely important to keep the optics clean and free from dust. Especially when using high energy puls lasers, dust on lenses can easily destroy lens coatings and substrates. Check the lenses regulary. Do not scratch or rub on lenses! If you see any cracks or scratches on the lenses, please contact **LaVision**.
Use compressed air to regularly remove dust particles from lenses and CCDs. You do not need to remove the lenses from the set up to do this.

3.1 Lenses and Filters

If lenses and filters are still dirty, you should clean them with ethanol. Do **not** use agressive solvents like gasolina or acetone and do not use a dry cloth to clean the lenses.

- Wear powder-free disposable gloves.
- Moisturize a lens cleaning tissue with pure ethanol. See figure 3.2.
- Wipe this tissue slowly, continuously and only once over the whole lens or filter surface so that the ethanol on the lens already evaporates close behind the tissue. By this method you will get a perfect stripe-free lens surface. See figure 3.3.
- Repeat this step if lens is not clean.
- Only carefully rub tissue or do rotary moves when there is persistent dirt. Afterwards, start from 1 to get a stripe free surface.

Try this method even with negative lenses.

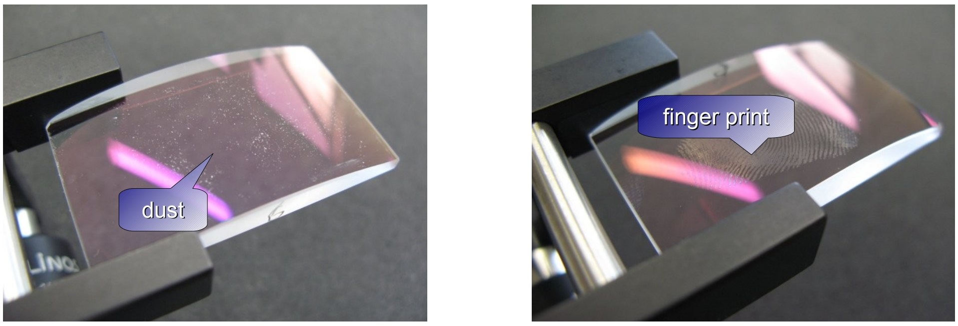

Figure 3.1: Lens dusty and greasy

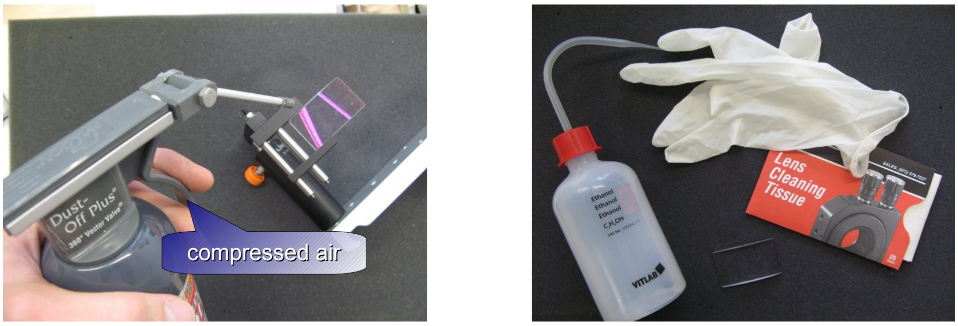

Figure 3.2: Use compressed air to remove dust from lenses. If that does not help use ethanol and lens cleaning tissues

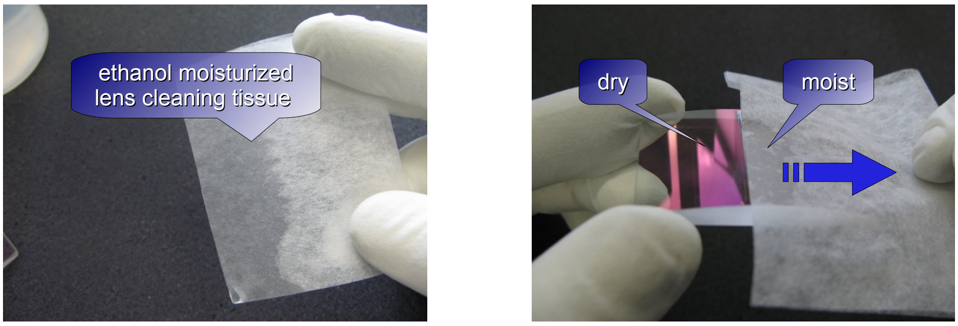

Figure 3.3: Moisturize the tissue. Slide it slowly over the lens so that it dries while sliding

3.2 CCDs

IF a CCD is still dirty after cleaning with compressed air **and** this is visible on the image, you should clean it with ethanol. Most cameras are equipped with an input window in front of the CCD. If your camera does not have an input window, be extremely careful as you have direct contact to the CCD glass itself. Be aware of what you do! Ask **LaVision** for help if you are not sure about this step.

Note that every time the input window of the CCD is cleaned, there is the possibility of surface damage. Do not clean the input window unless it is absolutely necessary!

Do **not** use agressive solvents like gasoline or acetone and do not use a dry cloth to clean the CCD.

- Unplug the camera from any power supply before cleaning it!
- Use a cotton swab dipped in pure ethanol and wipe only on the glass surface of the input window. Do not get any ethanol on the metallic parts such as the lens thread, because tiny detached particles may scratch the surface.

4 Support

If you have a technical problem or a question regarding hardware or software which is not adequately addressed in the documentation, please contact your local representative or **LaVision** service directly.

You can contact service at **LaVision** GmbH by:

e-mail: **service@lavision.de**
phone: **+49 551 9004 229**

Alternatively, you may submit your problem using the **Support Request Form** in the **Support** section of the **LaVision** website `www.lavision.com`.

In order to speed up your request, please include the following information:

- The order number of your system.
- The number of the used dongle (if available).
- Article number (if available).
- Serial number (if available).
- A short description of the problem.

4.1 Shipment of Defective Items

If any item needs to be returned to **LaVision** GmbH for service or repair, please contact the **LaVision** service to obtain a **RMA** (Return Material Authorization) number together with an RMA form. This will list all items with SN and a short description of the problem. Place the RMA form in the box with the item(s) being returned. Return the authorized item(s) according to the shipping instructions.

Shipping instructions:

- Be sure to obtain an RMA number and RMA form.
- Add the signed RMA form to the shipping documents.
- Ship only the items that are authorized.

- Use the original boxes to avoid damages during transportation.
- **Use antistatic bags for computer boards!**
- Ship returned items to:

  LaVision GmbH
  Anna-Vandenhoeck-Ring 19

  37081 Göttingen

  GERMANY

Note: Shipments received by **LaVision** without an RMA number may be refused.

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